Transport

Mark scheme and worked solutions for 25 questions: 8 multiple choice, 17 structured.

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1 · Part 4

Paper 1 & 2 Transport [1]

Answer: B

Worked solution

Small organisms have a large surface area to volume ratio. The surface is big enough for the amount of cytoplasm inside, and the distance from the membrane to the centre of the cell is very short, so diffusion is fast enough on its own.

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2 · Part 4

Paper 2 only Transport [1]

Answer: D

Explanation

The specification defines transpiration as the evaporation of water from the surface of a plant. A describes gas exchange for photosynthesis, B is a product of photosynthesis, and C is wrong because water moves up the plant in the xylem, not down in the phloem.

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3 · Part 4

Paper 2 only Transport [1]

Answer: A

Explanation

Moving air carries water vapour away from the leaf surface, keeping the concentration gradient of water vapour steep, so water vapour diffuses out faster. B (higher humidity) reduces the gradient and slows transpiration. C (lower temperature) slows evaporation and diffusion. D (lower light intensity) causes stomata to close, slowing transpiration.

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4 · Part 4

Paper 2 only Transport [1]

Answer: C

Explanation

Cutting and fitting the stem under water stops air entering the cut end. An air bubble in the stem would block the column of water so the shoot could not take up water and the readings would be wrong. A is wrong because the leaves still transpire; B and D have nothing to do with cutting the stem under water.

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5 · Part 4 Paper 1 & 2 Transport [1]

Answer: A

Explanation

Without a nucleus there is more room in the cell for haemoglobin, so each cell can carry more oxygen. B is wrong because cells without a nucleus cannot divide. C is wrong because more respiration by the red blood cell would use up oxygen, not help transport it. D is wrong because cell membranes are partially permeable and the nucleus has no effect on this.

6 · Part 4 Paper 1 & 2 Transport [1]

Answer: C

Explanation

The right ventricle pumps deoxygenated blood into the pulmonary artery, which takes it to the lungs. The aorta (A) leaves the left ventricle. The pulmonary vein (B) brings blood from the lungs into the left atrium. The vena cava (D) brings blood from the body into the right atrium.

7 · Part 4 Paper 1 & 2 Transport [1]

Answer: D

Explanation

Capillary walls are only one cell thick, so the diffusion distance between the blood and the body cells is very short. A describes arteries, B describes veins, and C is wrong because capillaries have a very narrow lumen.

8 · Part 4 Paper 1 & 2 Transport [1]

Answer: B

Explanation

The hepatic portal vein carries blood rich in digested food from the small intestine to the liver. The hepatic artery (A) carries oxygenated blood from the aorta to the liver. The hepatic vein (C) carries blood away from the liver. The vena cava (D) returns blood from the body to the heart.

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9 · Part 4

Paper 1 & 2 | Transport | [6]

Mark scheme

(a) **[3]**

- surface area = $6 \times 5^2 = 150$ (mm^2) (1)
- volume = $5^3 = 125$ (mm^3) (1)
- ratio = $150 \div 125 = 1.2$ (1)

ALLOW 1.2 : 1

ECF correct ratio from an incorrect surface area or volume

(b) **[3]**

Any three from:

- as the size (side length) increases, the surface area to volume ratio decreases (1)
- quoted data, e.g. ratio falls from 6.0 for P to 1.2 for R (1)
- a large organism has too little surface area for the volume of cells that need oxygen (for respiration) (1)
- the diffusion distance from the surface to cells in the centre is too long, so diffusion is too slow (1)
- so a transport system is needed to carry oxygen to the cells (1)

ECF data quoted from an incorrect answer to (a)

Worked solution

(a) For model R the side length is 5 mm. Surface area = $6 \times 5 \times 5 = 150$ mm^2. Volume = $5 \times 5 \times 5 = 125$ mm^3. Ratio = $150 \div 125 = 1.2$. Check: as the side length goes $1 \rightarrow 2 \rightarrow 5$, the ratio goes $6.0 \rightarrow 3.0 \rightarrow 1.2$; each ratio equals $6 \div$ side length, so $6 \div 5 = 1.2$.

(b) Bigger organisms have a smaller surface area to volume ratio. Each square millimetre of surface has to supply oxygen to a larger volume of respiring cells, and cells deep inside are a long way from the surface. Diffusion over long distances is far too slow, so large multicellular organisms need a transport system (such as blood) to carry oxygen close to every cell.

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10 · Part 4

Paper 1 & 2 Transport [6]

Mark scheme

(a) **[2]**

- as the distance increases, the time taken increases (1)
- the time increases much more than the distance: each 10 times increase in distance increases the time 100 times (1)

ALLOW time is not proportional to distance OR time increases faster than distance

(b) **[1]**

- 25 000 ÷ 3600 = 6.9 hours (1)

ALLOW 6.94 OR 7 hours

(c) **[3]**

Any three from:

- oxygen would take hours (about 7 hours or more) to diffuse to the cells deep inside the large animal (1)
- cells need a constant supply of oxygen for (aerobic) respiration, so they would die OR could not release enough energy (1)
- a transport system carries oxygen (in the blood) close to all cells so diffusion distances are short (1)
- in the single-celled organism, no part is more than about 0.1 mm from the surface, so diffusion takes only seconds (about 2.5 s or less) (1)

Worked solution

(a) Read down the table: 0.01 mm takes 0.025 s, 0.1 mm takes 2.5 s, 1 mm takes 250 s. Each time the distance is multiplied by 10, the time is multiplied by 100. So time rises much faster than distance.

(b) 25 000 s ÷ 3600 s per hour = 6.94 hours, about 6.9 hours. Check: 7 × 3600 = 25 200 s, just over 25 000 s.

(c) Cells need oxygen all the time for respiration. Diffusing 10 mm takes about 7 hours, which is far too slow to keep deep cells alive, so a large animal needs blood to carry oxygen close to every cell. A single cell 0.1 mm across is supplied by diffusion in a few seconds, so it does not need a transport system.

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11 · Part 4

Paper 1 & 2 Transport [6]

Mark scheme

(a) **[3]**

- (xylem) transports water (1)
- (xylem) transports mineral ions (1)
- from the roots to other parts of the plant OR up the stem to the leaves (1)

(b) **[1]**

- 24 ÷ 3 = 8 cm per hour (1)

(c) **[2]**

- sucrose (1)
- amino acids (1)

ALLOW sugar for sucrose

REJECT glucose OR starch

Worked solution

(a) Xylem carries water and dissolved mineral ions from the roots up the stem to the leaves and other parts of the plant. The dye moved in the water, which is why it showed up in the xylem and then in the leaf veins.

(b) The dye travelled 24 cm in 3 hours, so rate = 24 ÷ 3 = 8 cm per hour. Check: 8 cm per hour × 3 hours = 24 cm.

(c) Phloem carries sucrose and amino acids made in the leaves to other parts of the plant.

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12 · Part 4

Paper 1 & 2 Transport [6]

Mark scheme

(a) **[2]**

- the xylem is still intact OR was not removed (1)
- so water (and mineral ions) can still be transported from the roots to the leaves (1)

(b) **[2]**

- phloem transports sucrose (and amino acids) from the leaves down to other parts of the plant (1)
- the ring blocks this transport so sucrose (and amino acids) build up just above the ring (1)

ALLOW the build-up is used for extra growth just above the ring

(c) **[2]**

- sucrose (and amino acids) from the leaves can no longer reach the roots (1)
- so the root cells have less sucrose for respiration OR less amino acids for making proteins for growth (1)

Worked solution

Removing the bark removes the phloem but leaves the xylem.

(a) Water travels up from the roots in the xylem, which is undamaged, so the leaves keep getting water and do not wilt.

(b) Phloem carries sucrose and amino acids from the leaves to other parts of the plant, including down to the roots. At the ring this flow stops, so these substances collect above the ring and the tissue there swells.

(c) The roots depend on the leaves for sucrose (used in respiration to release energy) and amino acids (used to make proteins). With the phloem cut, the roots are starved of these, so they grow slowly.

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13 · Part 4 Paper 2 only Transport [7]

Mark scheme

(a) **[3]**

- by osmosis (1)
- water moves from the soil, where it is a more dilute solution (higher concentration of water), into the cell, a more concentrated solution (lower concentration of water) (1)
- through the partially permeable (cell) membrane (1)

ALLOW down a concentration gradient of water for the second point

(b) **[2]**

- the long, thin extension gives a large surface area (1)
- so more water can be absorbed (in a given time) OR water is absorbed faster (1)

(c) **[2]**

- water now leaves the root hair cells (by osmosis) OR the root hair cells can no longer absorb water (1)
- so less water travels to the leaves (in the xylem) and the leaf cells lose water, so the plant wilts (1)

Worked solution

(a) Osmosis is the movement of water from a dilute solution to a more concentrated solution through a partially permeable membrane. Soil water is more dilute than the solution in the root hair cell's vacuole, so water moves into the cell.

(b) The long, thin shape gives a large surface area in contact with soil water, so more water can enter by osmosis at the same time.

(c) Seawater makes the soil solution more concentrated than the cell contents. Now the water concentration gradient is reversed: water leaves the root hair cells instead of entering them. The leaves stop receiving enough water through the xylem, so they wilt.

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14 · Part 4 Paper 2 only Transport [7]

Mark scheme

(a) **[1]**

- to stop water evaporating from the water surface, so any loss in mass is due to the shoot only OR due to transpiration only (1)

(b) **[2]**

- loss in mass = 51.4 − 46.8 = 4.6 (g) (1)
- 4.6 ÷ 51.4 × 100 = 8.9 (%) (1)

ECF correct percentage from an incorrect loss in mass

(c) **[2]**

- at higher temperature water evaporates faster (from cells inside the leaf) OR water molecules have more kinetic energy (1)
- water vapour diffuses out of the leaf (through the stomata) faster, so the rate of transpiration is higher (1)

(d) **[2]**

- the air around Z is very humid OR contains a lot of water vapour (1)
- so the concentration gradient of water vapour between the inside of the leaf and the air is small, so less water vapour diffuses out (1)

Worked solution

(a) Without the oil, water would evaporate straight from the water surface and add to the mass loss, so we could not tell how much water the shoot lost.

(b) Shoot X lost 51.4 − 46.8 = 4.6 g. Percentage loss = 4.6 ÷ 51.4 × 100 = 8.95%, which is 8.9% to 2 significant figures. Check: 9% of 51.4 g is about 4.6 g.

(c) Transpiration is evaporation of water from the plant surface. At 30 °C water evaporates and diffuses faster than at 20 °C, so Y lost more water.

(d) The wet paper in the closed box makes the air humid. Water vapour leaves the leaf by diffusion, which depends on a concentration gradient; with very humid air the gradient is small, so Z lost very little water.

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15 · Part 4 Paper 2 only Transport [9]

Mark scheme

(a) **[1]**

- (30 + 33 + 31) ÷ 3 = 31 (mm) (1)

ALLOW 31.3

(b) **[2]**

- 42 ÷ 5 (1)
- = 8.4 mm per minute (1)

(c) **[2]**

Any two from:

- temperature (1)
- humidity (1)
- wind speed OR air movement (1)
- the same leafy shoot OR same number of leaves OR same leaf area (1)
- the same time for each measurement (5 minutes) (1)

IGNORE amount of water

(d) **[3]**

- more light makes the stomata open (wider) (1)
- so more water vapour diffuses out of the leaves OR the rate of transpiration increases (1)
- more water is drawn up the xylem to replace the water lost, so the shoot takes up water faster (1)

ALLOW the lamp warms the leaves so water evaporates faster, for the second point

(e) **[1]**

- place a heat shield OR a glass tank of water between the lamp and the shoot (1)

ALLOW use an LED lamp OR a lamp that gives out little heat

ALLOW measure the temperature near the shoot with a thermometer and keep it the same

Worked solution

(a) Add the three trials and divide by 3: 94 ÷ 3 = 31.3, so 31 mm to match the other means.

(b) In 5 minutes the bubble moved a mean of 42 mm, so rate = 42 ÷ 5 = 8.4 mm per minute. Check: 8.4 × 5 = 42.

(c) Only light intensity should change. Temperature, humidity and air movement all change the rate of transpiration, so they must be kept the same, and the same shoot should be used throughout.

(d) In brighter light the stomata open wider. More water vapour diffuses out of the leaf, so more water is pulled up the xylem from the potometer, and the bubble moves further.

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(e) A lamp close to the shoot also warms it, and higher temperature increases transpiration on its own. A heat shield such as a tank of water between lamp and shoot absorbs the heat but lets the light through.

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16 · Part 4

Paper 1 & 2 Transport [7]

Mark scheme

(a) **[1]**

- plasma (1)

(b) **[2]**

- heat (energy) from the muscles is absorbed by the plasma OR warms the plasma (1)
- the plasma carries the heat in the blood to the skin, where it is lost to the surroundings (1)

(c) **[4]**

- urea (1)
- carbon dioxide (1)
- hormones (1)
- digested food OR glucose OR amino acids (1)

ALLOW CO_2 for carbon dioxide; a named hormone, e.g. insulin OR adrenaline

REJECT starch OR protein for the last row

Worked solution

(a) The liquid part of the blood is plasma; it makes up just over half of the blood volume. The cells (red blood cells, white blood cells and platelets) settle below it.

(b) Respiring muscles release heat. The plasma absorbs this heat and the circulating blood carries it to the skin, where it is lost to the air.

(c) Plasma carries dissolved substances: urea from the liver to the kidneys, carbon dioxide from respiring cells to the lungs, hormones from glands to target organs, and digested food such as glucose and amino acids from the small intestine to the liver.

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17 · Part 4

Paper 1 & 2 Transport [7]

Mark scheme

(a) **[2]**

- (6.2 − 5.0) ÷ 5.0 × 100 (1)
- = 24 (%) (1)

(b) **[2]**

- each breath at 4500 m provides less oxygen OR less oxygen is absorbed into the blood (1)
- more red blood cells (more haemoglobin) can carry more oxygen to body cells for respiration (1)

(c) **[3]**

- the biconcave shape gives a large surface area (to volume ratio) (1)
- so oxygen can diffuse into and out of the cell quickly (1)
- haemoglobin combines with oxygen (in the lungs) and releases it (in the tissues) (1)

ALLOW haemoglobin forms oxyhaemoglobin

Worked solution

(a) Increase = $6.2 \times 10^6 - 5.0 \times 10^6 = 1.2 \times 10^6$. Percentage increase = 1.2 ÷ 5.0 × 100 = 24%. Check: 5.0 × 1.24 = 6.2.

(b) At high altitude each breath brings in less oxygen. Having more red blood cells, and so more haemoglobin, lets the blood carry more oxygen so cells get enough for respiration.

(c) The biconcave disc shape gives a large surface area for oxygen to diffuse in and out quickly. Haemoglobin combines with oxygen in the lungs and releases it in the tissues.

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18 · Part 4 Paper 1 & 2 Transport [6]

Mark scheme

(a) **[3]**

- phagocytes move to the bacteria OR are attracted to the bacteria (1)
- they ingest OR engulf the bacteria (1)
- they digest the bacteria (using enzymes) (1)

(b) **[2]**

- antibodies are specific to one type of pathogen (1)
- a different bacterium has a different shape OR different antigens, so the antibodies cannot attach to it (1)

(c) **[1]**

- more white blood cells were made to fight OR destroy the bacteria (infection) (1)

Worked solution

(a) Phagocytes are attracted to the bacteria, surround them and take them into the cell (ingest them), then digest them with enzymes.

(b) Lymphocytes make antibodies that fit only the pathogen that caused the response. A different bacterium has a different shape on its surface, so these antibodies cannot bind to it.

(c) The body is fighting an infection, so it makes more white blood cells (phagocytes and lymphocytes) to destroy the bacteria.

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19 · Part 4 Paper 2 only Transport [7]

Mark scheme

(a) **[2]**

- after the infection antibodies were produced sooner OR faster, e.g. 40 units 3 days after the infection compared with only 2 units 5 days after the vaccine OR the peak was reached 10 days after the infection compared with 15 days after the vaccine (1)
- after the infection a greater quantity (concentration) of antibodies was produced, e.g. peak of 200 compared with 18 (arbitrary units) (1)

ALLOW antibody concentration stayed higher for longer after the infection

(b) **[2]**

- 200 ÷ 18 (1)
- = 11 (times) (1)

ALLOW 11.1

(c) **[3]**

Any three from:

- the vaccine stimulated lymphocytes to produce memory cells (1)
- memory cells remain in the blood after the first response (1)
- on infection, memory cells recognise the pathogen (antigen) quickly (1)
- memory cells (divide quickly to) produce lymphocytes that release antibodies sooner, faster and in greater quantity (1)

ALLOW so the pathogen is destroyed before it causes illness

Worked solution

(a) After the vaccine, antibodies rose slowly to a peak of only 18 units at day 15. After infection on day 60, antibodies rose to 150 units within 6 days and peaked at 200 units on day 70. So the second response was sooner, faster and much bigger.

(b) Highest after infection = 200; highest after vaccination = 18. 200 ÷ 18 = 11.1, so about 11 times greater. Check: 18 × 11 = 198, close to 200.

(c) The vaccine caused lymphocytes to make memory cells. These stay in the blood. When the real pathogen arrives, the memory cells recognise it and quickly produce lymphocytes that release large amounts of antibody, so the pathogen is destroyed before the person becomes ill.

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20 · Part 4 Paper 2 only Transport [6]

Mark scheme

(a) **[2]**

- platelets are involved in blood clotting, so with fewer platelets a clot forms more slowly OR less clotting happens (1)
- the cut is not sealed, so blood continues to be lost (1)

(b) **[2]**

- a clot seals the cut (1)
- without a clot, micro-organisms OR pathogens can enter the body through the cut (1)

(c) **[2]**

- (110 000 − 20 000) ÷ 20 000 × 100 (1)
- = 450 (%) (1)

Worked solution

(a) Platelets help blood to clot. A clot plugs the cut and stops bleeding. With far fewer platelets, the clot forms slowly or not at all, so the cut keeps bleeding.

(b) A clot also forms a barrier that stops micro-organisms getting into the body. If clotting is slow, the cut stays open for longer and pathogens can enter, so infection is more likely.

(c) Increase = 110 000 − 20 000 = 90 000. Percentage increase = 90 000 ÷ 20 000 × 100 = 450%. Check: 20 000 × 5.5 = 110 000, which is a 450% increase.

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21 · Part 4 Paper 1 & 2 Transport [8]

Mark scheme

(a) **[2]**

- the septum stops oxygenated blood mixing with deoxygenated blood (1)
- so the blood pumped to the body carries as much oxygen as possible (1)

ALLOW keeps blood in the left side separate from blood in the right side, for the first point

(b) **[2]**

- the left ventricle pumps blood (through the aorta) to the whole body, but the right ventricle pumps blood only to the lungs (1)
- the left ventricle needs more muscle to contract more strongly and produce a higher pressure (1)

(c) **[4]**

Any four from:

- blood enters the atria from veins (vena cava and pulmonary vein) (1)
- the atria contract, pushing blood into the ventricles (1)
- the ventricles contract, pushing blood out into the arteries (aorta and pulmonary artery) (1)
- valves between the atria and ventricles close when the ventricles contract, preventing backflow into the atria (1)
- valves at the base of the arteries open as blood leaves, then close to stop blood flowing back into the ventricles (1)
- valves make sure blood flows in one direction only through the heart (1)

ALLOW tricuspid OR bicuspid valves for valves between atria and ventricles; semilunar valves for valves at the base of the arteries

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Worked solution

(a) The left side of the heart contains oxygenated blood from the lungs and the right side contains deoxygenated blood from the body. The septum keeps them apart so the blood sent to the body has as much oxygen as possible.

(b) The left ventricle must push blood all the way round the body, so it needs thicker muscle to produce a higher pressure. The right ventricle only pumps blood to the nearby lungs.

(c) Blood flows into the atria from the veins. The atria contract and push blood into the ventricles. The ventricles then contract; the valves between atria and ventricles close so blood cannot flow back, and blood is forced out into the aorta and pulmonary artery. Valves at the start of these arteries close afterwards so blood cannot flow back into the ventricles. The valves keep blood flowing one way.

22 · Part 4

Paper 1 & 2 Transport [8]

Mark scheme

(a) **[2]**

- (150 − 68) ÷ 68 × 100 (1)
- = 121 (%) (1)

ALLOW 120.6

(b) **[4]**

Any four from:

- muscles contract more, so they respire faster (1)
- muscles need more oxygen (and glucose) for aerobic respiration (1)
- more carbon dioxide is produced and must be removed (1)
- a faster heart rate pumps more blood to the muscles per minute (1)
- so oxygen and glucose are delivered faster and carbon dioxide is removed faster (1)

ALLOW an increase in carbon dioxide in the blood is detected and the heart rate is increased

(c) **[2]**

- (nervousness or excitement causes) adrenaline to be released (into the blood) (1)
- adrenaline increases heart rate, preparing the body for action OR to supply more oxygen and glucose to the muscles (1)

Worked solution

(a) Increase = 150 − 68 = 82 beats per minute. Percentage increase = 82 ÷ 68 × 100 = 120.6%, which is 121% to 3 significant figures. Check: 68 × 2.21 = 150.3.

(b) During running the leg muscles contract much more and need more energy, so they respire faster. This needs more oxygen and glucose and produces more carbon dioxide. The heart beats faster so more blood flows to the muscles each minute, delivering oxygen and glucose and removing carbon dioxide.

(c) Being nervous or excited makes the body release adrenaline. Adrenaline increases heart rate, getting the body ready for action by supplying the muscles with more oxygen and glucose.

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23 · Part 4 Paper 1 & 2 Transport [8]

Mark scheme

(a) **[3]**

- smoking increases the risk of dying from CHD, e.g. 120 to 300 for men with normal blood pressure (1)
- high blood pressure increases the risk, e.g. 120 to 260 for non-smokers (1)
- men who smoke and have high blood pressure have the highest risk (680), more than either factor alone (1)

ALLOW smoking increases risk more than high blood pressure (300 compared with 260)

ALLOW smokers with high blood pressure are about 5.7 times more likely to die than non-smokers with normal blood pressure

(b) **[3]**

Any three from:

- coronary arteries supply the heart muscle with blood (1)
- fatty material narrows OR blocks the coronary artery, so less blood flows through it (1)
- the heart muscle gets less oxygen (and glucose) (1)
- so heart muscle cells cannot respire (aerobically) OR die, and the heart muscle stops contracting properly (1)

ALLOW a blood clot forms at the fatty deposit and blocks the artery

(c) **[2]**

Any two from:

- diet high in (saturated) fat OR cholesterol (1)
- lack of exercise (1)
- obesity OR being overweight (1)
- stress (1)
- genetic factors OR family history (1)
- diet high in salt (1)

Worked solution

(a) Compare each group with the non-smokers with normal blood pressure (120). Smoking alone raises deaths to 300; high blood pressure alone raises them to 260. With both, the figure is 680, the highest of all and more than either factor causes on its own.

(b) The coronary arteries carry blood to the heart muscle itself. Fatty material makes the lumen narrower, so less blood, and less oxygen, reaches part of the heart muscle. Without oxygen these cells cannot respire aerobically and they die, so that part of the heart stops working: a heart attack.

(c) Other risk factors include a diet high in saturated fat, lack of exercise, being overweight, stress and inherited (genetic) factors.

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24 · Part 4 Paper 1 & 2 Transport [6]

Mark scheme

(a) **[3]**

- arteries carry blood away from the heart at high pressure (1)
- the thick wall (with muscle) withstands the high pressure without bursting (1)
- the elastic tissue stretches and recoils, keeping the blood flowing OR smoothing out the pressure (1)

(b) **[2]**

- blood in veins is at low pressure (1)
- valves prevent blood flowing backwards OR keep blood flowing towards the heart (1)

(c) **[1]**

- less resistance to blood flow, so blood at low pressure can return to the heart easily (1)

ALLOW veins can hold a large volume of blood

Worked solution

(a) Each heartbeat pushes blood into the arteries at high pressure. Thick walls with muscle stop the arteries bursting, and elastic tissue stretches as blood surges in and then recoils, which keeps blood moving.

(b) By the time blood reaches the veins its pressure is low, so it could flow backwards. Valves close if blood starts to flow back, keeping it moving towards the heart.

(c) A wide lumen gives less resistance, so low-pressure blood can still flow back to the heart easily.

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25 · Part 4 Paper 1 & 2 Transport [6]

Mark scheme

(a) **[3]**

- pulmonary vein (1)
- left atrium then left ventricle (1)
- aorta then renal artery (1)
- each mark only if in the correct position in the sequence

IGNORE valves and extra correct structures named in the correct position, e.g. bicuspid valve

(b) **[1]**

- renal vein (1)

(c) **[2]**

- in the lungs, oxygen diffuses from the air in the alveoli into the blood (1)
- blood arriving at the lungs has come from the body, where cells have used oxygen for respiration (1)

Worked solution

(a) Blood from the lungs returns to the heart in the pulmonary vein and enters the left atrium. It passes into the left ventricle, which pumps it into the aorta. A branch of the aorta, the renal artery, takes it to the kidney.

(b) Blood leaves the kidney in the renal vein, which joins the vena cava.

(c) Blood arriving at the lungs has just been round the body, where cells used oxygen for respiration. In the lungs, oxygen diffuses from the alveoli into the blood, so blood leaving the lungs is rich in oxygen.

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